MIL-P-14585A(CmlC)
26 April 1962
SUPERSEDING
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23 July 1957

MILITARY SPECIFICATION

PROJECTILE LOAD, 6 INCH 47 CALIBER, WP, MK 13; FILLING ASSEMBLY

1. SCOPE

1.1 This specification covers the assembly and filling of a projectile load, without explosive elements, with white phosphorus (WP).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320 - Fiberboard; Sheet, Stock and Cut Shapes Standards.

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

FSC 1320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

DEPARTMENT OF THE NAVY - BUREAU OF NAVAL WEAPONS

289025 - Projectile, Smoke, WP, 6"/47, MK 41 List of Drawings, Subassemblies and Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-90.")

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 289025 or on the subsidiary drawings, used in the filling and assembly of the projectile load, shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile load shall conform to the specifications and drawings listed on Drawing 289025 and subsidiary drawings.

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3.2 Filling and assembly.

3.2.1 Filling. With the closure cap, VCI paper, and foreign matter removed, the projectile load shall be filled with molten white phosphorus conforming to Specification MIL-C-215 (see 6.2). Whenever process water is used in WP filling lines, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The filling shall be solidified in a nose-up position.

3.2.2 Filling height. The surface of the WP in the filled and assembled projectile load shall be between 1.86 and 2.36 inches below the top of the casing, and the depth of water (if used) on top of the WP shall be no more than 1/4 inch. All readings shall be taken when the projectile load is in a vertical nose-up position in accordance with 4.6.1.

3.2.3 Assembly. The projectile load shall be assembled as shown on Drawing 1380988, which is listed on Drawing 289025.

3.3 Cleaning after closing. The exterior of the assembled projectile load and the interior of the burster well shall be thoroughly cleaned to remove any oil, grease, and WP contamination (see 6.3).

3.4 Leakage. The cleaned projectile load shall not leak when tested in accordance with 4.6.2.

3.5 Workmanship. The projectile load shall be free from foreign matter (oil or grease) and damage such as dents, cracks, or deformations which would impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

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4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the projectile loads produced by one manufacturer, at one plant, under the same filling conditions and offered for delivery at one time. However, no more than one lot of casings and burster wells shall be represented in any one lot of completed projectile loads.

4.4 Sampling. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.5 Examination. Sample projectile loads shall be examined in accordance with the classification of defects and with Standard MIL-STD-105.

4.5.1 Classification of defects.

4.5.1.1 Projectile load, 6"/47, WP, MK 13, assembly and marking diagram (Dwg 1380988).

Categories	Defects	Acceptance standards
*Critical:		
1	Leakage	4.6.2
2	WP contamination (see 3.3)	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Marking incorrect	
103	Filling excessive	4.6.1
104	Water excessive	4.6.1
105	Damage (see 3.5)	
Minor:	AQL 2.5 percent defective	
201	Filling insufficient	4.6.1
202	Protective finish missing	
203	Marking illegible	

*Each projectile load in each lot shall be examined for these defects.

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4.5.1.2 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Container incorrect
102	Filler pads missing or insufficient to make a tight pack
103	Container closure incorrect
104	Strapping loose or missing
105	Marking incorrect, missing or illegible

4.6 Tests.

4.6.1 Filling height. The assembled projectile load shall be radiographed (see 6.1) when the temperature of the WP filling is 130° ± 10°F using Chemical Corps Inspection Standard Number R13-31.

4.6.2 Leakage. Subsequent to the final cleaning operation and while the filling is still molten (130° ± 10°F), the assembled projectile load shall be placed in a nose-down position or rolled on its side for a minimum period of 1 minute. The emission of smoke shall constitute leakage.

5. PREPARATION FOR DELIVERY

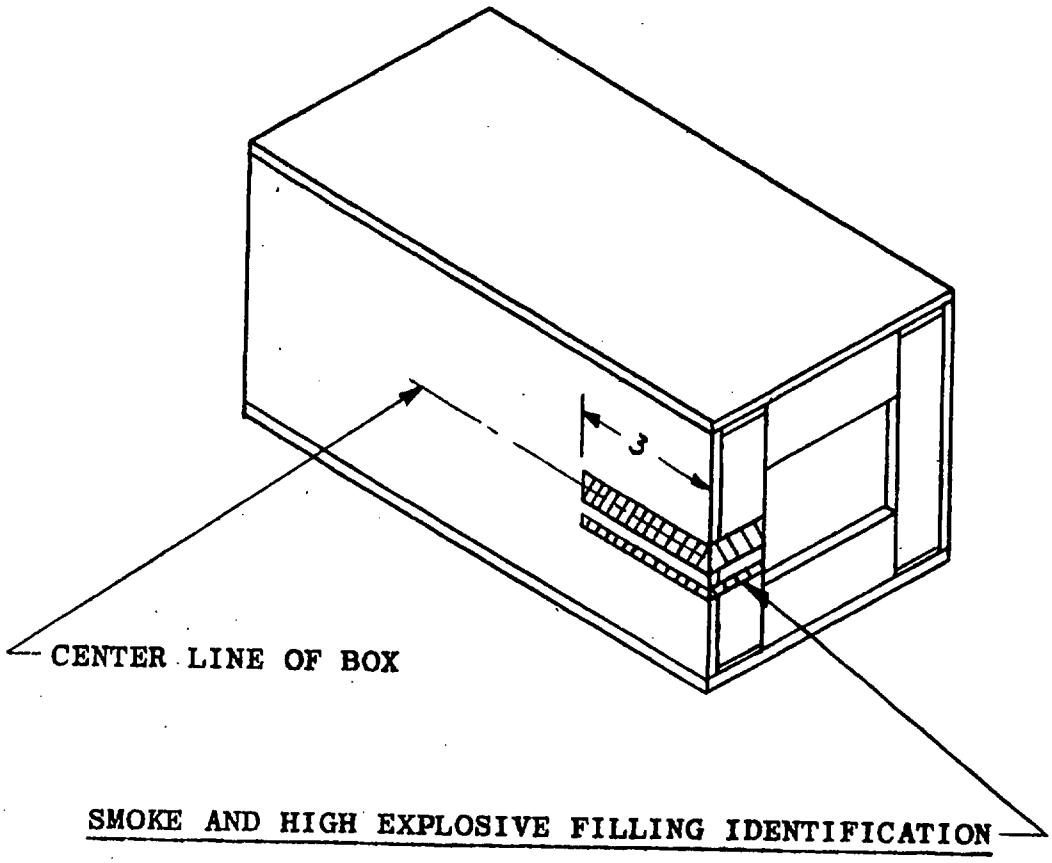
5.1 Packing.

5.1.1 Level A. The filled projectile loads shall be packed not to exceed the gross weight of 150 pounds per container in class 2, style optional, nailed wood boxes, conforming to Specification PPP-B-621. Filler pads, fabricated from type CF or SF, class-weather resistant fiberboard, conforming to Specification PPP-F-320, shall be placed between the sides, ends, top, and bottom of the box and the projectile loads. Additional filler pads shall be used as required to make a tight pack. The box shall be closed in conformance with applicable requirements of Specification PPP-B-621, with strapping to be in conformance with the appendix thereto. Galvanized strapping shall be used.

5.1.2 Level B. The filled projectile loads shall be packed as specified in 5.1.1, except that class 1 nailed wood boxes and coated-finish strapping shall be used.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with ICC regulations and Standard MIL-STD-129, except that color code filling identification marking shall be as shown on figure 1.

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SMOKE AND HIGH EXPLOSIVE FILLING IDENTIFICATION

MARK CORNER OF BOX AS SHOWN, A 1/2 INCH WIDE, COLOR LIGHT GREEN NO. 24558 OR 34558 STRIPE, AND A 1/4 INCH WIDE, COLOR YELLOW NO. 23538 OR 33538 STRIPE, SPACED 1/2 INCH APART.

FIGURE I

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The supplier will not be required to furnish radiographic equipment and the inspection standard.
(c) The level of packing required.

6.2 Precautions. Because of the hazardous nature of filling munitions with WP, filling installations are advised to take all necessary precautions to protect personnel from injury while filling in accordance with this specification.

6.3 Decontamination. Projectile loads may be decontaminated by passing them through a hot water and detergent spray washer or by flushing them with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

6.4 Safety classification. This item caries the following safety classification: chemical group C, explosive hazard class 2 (Ref: Chemical Corps Safety Directive 385-10).

6.5 Filling aid. Prior to inserting the burster well, projectile loads filled with the surface of the WP 4.31 to 4.81 inches below the top of the casing and the depth of water (if used) on top of the WP 1/8 to 1/4 inch have been found to approximate the requirement in 3.2.2. The WP in the load will weigh approximately 10 3/4 pounds. This information is advisory only and is not intended as a criterion for filling.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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